

MAPQUEST III: Target Certification

When Reaching T Becomes a Verified Proof

The Logical Climax of the Hilbert–Tenneson MAPQUEST Program

Brian Tenneson

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Abstract

The first two MAPQUEST papers develop a target-directed automated theorem prover whose candidate inference spaces are explicitly addressable and may be traversed by rule-specific Hilbert space-filling schedules. Paper I develops the navigation architecture. Paper II adds Hilbert-fair revisitation, backward target cones, waiting inference frames, and a positive-completeness result showing that an existing finite proof cannot be permanently missed under effective and fair search assumptions. A logical question nevertheless remains: these are theorems about the search engine, not yet a proof of the current target. The present paper closes that gap. It begins by reviewing the first two papers, identifying what is still missing, and reminding the reader that the ATP—or, equivalently in the AMLD formulation, the theorem-proving logical decider—has one mathematical objective: prove the fixed target T . The search may use Hilbert traversal, Tenneson1, Professor guidance, Dreamer, four oracle interfaces, Depth Charge, Sentinel, Tenneson2, and a verified BANK. None of those devices is itself the proof. The climax is the MAPQUEST Target-Certification Theorem: under mild provenance, finiteness, and verifier-soundness assumptions, if T enters the verifier-gated BANK after a finite MAPQUEST run, then a finite certificate can be extracted and independently checked, hence $\Gamma \vdash_{\mathfrak{F}} T$. The final question is therefore answered exactly: when MAPQUEST says it reached T , did it actually prove T ?

The object of the machine is not to explore proof space.

It is to prove the fixed target T :

$$\boxed{\Gamma \vdash_{\mathfrak{F}} T.}$$

1 The three-paper arc

The MAPQUEST program is easiest to understand as a sequence of three questions.

1. How can proof search be navigated geometrically and algorithmically?
2. How can that navigation remain complete enough not to lose a finite proof route?
3. When the machine claims that it has reached the target, what turns that event into an actual proof of the target?

The first paper answers the first question. The second answers the second. This paper is devoted to the third.

1.1 Paper I

Paper I is titled

MAPQUEST for Formal Proof: A Hilbert–Tenneson Navigation Algorithm for Verifier-Gated Automated Theorem Proving.

It develops the core navigation architecture. Well-formed formulas are treated as zero-dimensional logical locations with exact identifiers and rational addresses. Rule applications, rather than formulas themselves, carry intrinsic geometric dimension: a rule with k ordered premises induces a k -dimensional premise-address component. Heterogeneous rule components are combined as a tagged coproduct. Hilbert traversals organize the candidate-premise space, while additional controller mechanisms decide which regions receive attention.

The fixed target is

$$T.$$

Its address identifies the destination, but the address difference itself is not proof distance. Accordingly, Paper I introduces a separate target-conditioned navigation layer: settlement-distance estimates, target structure, partial-credit guidance, and related control information determine which candidate inference regions appear useful.

Paper I therefore supplies the *road atlas and navigator*.

1.2 Paper II

Paper II is titled

MAPQUEST: Hilbert-Fair Navigation – From Addressable Inferences to a Verified Target: Positive Completeness, Target Cones, and Near-Target Guidance.

It addresses a weakness that remains after a single target-guided Hilbert pass. A legal inference can be encountered before its premises are in the verified BANK. If that address is never revisited, a genuine finite proof could be lost. Paper II therefore adds fair recurrence across rule components, address resolution, witness codes, and later BANK states.

Using explicit finite addresses for finite inference candidates and a fair scheduler, Paper II establishes the theorem-shaped guarantee called the

Hilbert-Fair Positive Completeness Theorem.

In schematic form,

$$\boxed{\Gamma \vdash_{\mathfrak{F}} T \implies \text{MAPQUEST eventually reaches a verified derivation of } T,}$$

provided the declared effective-enumerability, verifier, monotonicity, and fairness assumptions hold.

Paper II also develops the backward target cone. The machine does not only ask what follows from the current BANK. It also asks which legal inference patterns could conclude T , which premises those patterns require, which of those premises remain missing, and which missing premise should become the next temporary subtarget.

Thus Paper II supplies the *fairness backbone and target cone*.

2 What is still missing from the theory?

The first two papers still leave a logical gap.

The implication

$$\Gamma \vdash T \implies \text{MAPQUEST eventually finds a proof}$$

is a theorem about the behavior of MAPQUEST *assuming that a proof exists*. It does not itself prove the target currently presented to the machine.

That is not the ATP's objective. The ATP's objective is the opposite-direction event produced by an actual run:

$$\boxed{\text{MAPQUEST has reached } T \implies \Gamma \vdash T.}$$

The missing theory must therefore explain three things:

1. why every accepted BANK deposit remains formally justified;
2. why the provenance of a finite run ending at T can be assembled into a finite certificate;
3. why acceptance of that certificate entails the desired mathematical conclusion $\Gamma \vdash T$.

This is a certification problem, not a search-order problem.

3 Reminder: the ATP, or AMLD, is trying to prove T

Let

$$\mathfrak{F} = (\Sigma, \Gamma, \mathcal{R}, V)$$

be the frozen formal environment, with syntax Σ , hypotheses Γ , inference rules $\mathcal{R}$, and verifier V .

The machine may be called an automated theorem prover. In the AMLD formulation it may be regarded as the theorem-proving role of an automated mathematical logical decider. For the present paper the distinction is operationally irrelevant: its fixed mathematical task is

$$\boxed{\text{construct a finite verifier-accepted derivation of } T.}$$

The desired endpoint is not a favorable coordinate, a high Professor score, or a promising Dreamer branch. It is

$$\boxed{V(\Gamma, T, \Pi_T) = 1}$$

for some finite certificate Π_T , and hence

$$\boxed{\Gamma \vdash_{\mathfrak{F}} T.}$$

4 The tools are subordinate to the target

MAPQUEST may use many mechanisms:

- Hilbert traversal : locality-preserving organization of candidate rule applications,
- Tenneson1 : target-directed route selection and settlement control,
- Professor : throttled partial-credit guidance,
- Dreamer : counterfactual route, representation, and lemma proposals,
- O_1, O_2, O_3, O_4 : role, resource, strategy, and certificate guidance,
- Depth Charge : bounded escape from a stagnant search basin,
- Sentinel : resource and execution governance,
- Tenneson2 : sealed, reproducible, verifier-governed execution,
- BANK : persistent verified mathematical state.

All of these devices serve one end:

$$\Gamma \rightsquigarrow C_1 \rightsquigarrow C_2 \rightsquigarrow \dots \rightsquigarrow T.$$

4.1 What the Hilbert curves contribute

For a rule r of arity k , MAPQUEST can organize ordered premise tuples in a k -dimensional address component and traverse finite approximations of that component by a discrete Hilbert order. The curve can therefore run in parallel with the inference-rule geometry: it is an ordering of candidate rule applications in the corresponding premise space.

A crucial qualification is required. A Hilbert curve does not automatically make every candidate inference valid. Instead, MAPQUEST uses verifier gating to restrict the *certified realized path* to accepted transitions. If

$$c_i = (r_i; A_{i,1}, \dots, A_{i,k_i}; \xi_i)$$

is a realized proof step, then the accepted path satisfies

$$\boxed{V(c_i) = 1} \quad \text{for every certified step } c_i.$$

Thus the search geometry may range broadly, while the proof path preserved in the BANK consists entirely of verifier-value-1 transitions.

This is part of the value of Hilbert-guided verifier-gated navigation: the geometric traversal can parallel the inference components, while the trusted path remains inside the verifier-accepted portion of that search.

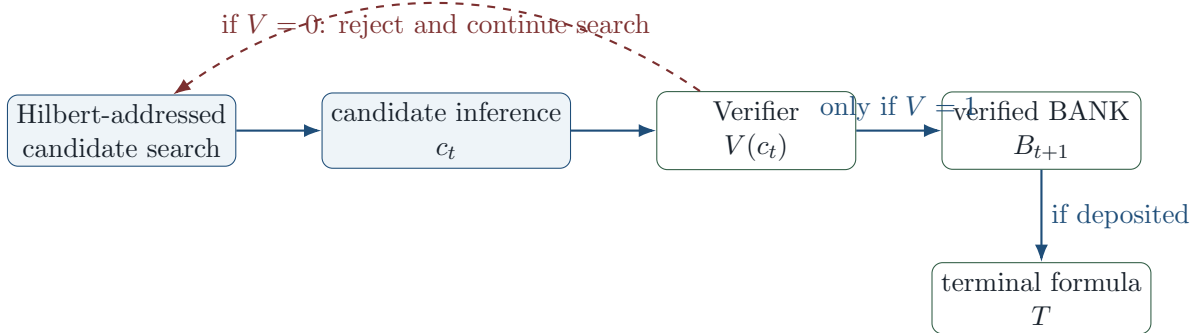


Figure 1: Search may be geometric and heuristic; trusted mathematical progress is verifier-gated.

5 The named results before the climax

Three named statements organize the theory.

1. **Hilbert-Fair Positive Completeness Theorem** (Paper II): if a finite proof of T exists and the declared fairness/effectivity assumptions hold, MAPQUEST does not permanently miss that proof.
2. **Verifier-Gated BANK Invariant** (proved below): every formula in the trusted BANK is derivable from the initial formal environment.

3. **MAPQUEST Target-Certification Theorem** (the climax): if a finite MAPQUEST run reaches T through the verifier-gated BANK, then the recorded provenance extracts to a finite certificate accepted by the verifier; therefore $\Gamma \vdash T$.

The first theorem concerns search completeness. The second concerns trusted-state preservation. The third finally concerns the actual target.

6 Verifier-gated BANK invariant

Let the initial BANK be

$$B_0 = \Gamma \cup \text{Ax}(\mathfrak{F}).$$

Assume BANK growth is monotone and verifier-gated. At stage t , a new conclusion C_t may be deposited only if there exists a finite candidate inference

$$\pi_t : A_{t,1}, \dots, A_{t,k} \Longrightarrow C_t$$

whose premises already occur in B_t , whose side conditions are satisfied, and for which

$$V(\pi_t) = 1.$$

Theorem 1 (Verifier-Gated BANK Invariant). *Assume V is sound for the declared formal system and certificate language. Then for every finite stage t ,*

$$B_t \subseteq \text{Th}_{\mathfrak{F}}(\Gamma).$$

Equivalently, every formula in the trusted BANK is derivable from the initial hypotheses and axioms of $\mathfrak{F}$.

Proof. By induction on t . Every element of B_0 is either an initial hypothesis or an axiom. Assume every element of B_t is derivable. If no new formula is deposited, the claim is immediate. Otherwise C_t is admitted only through a verifier-accepted legal inference whose premises $A_{t,1}, \dots, A_{t,k}$ already belong to B_t . By the induction hypothesis those premises are derivable from Γ , and by soundness of the verified rule application, so is C_t . Hence every element of B_{t+1} is derivable. Finite induction proves the claim for every t . $\square$

Professor, Dreamer, the four oracles, Hilbert navigation, Tenneson1, and Depth Charge may change *which proposal is tried*. They do not alter this invariant because none of them is permitted to deposit a statement directly into the trusted BANK.

7 Finite provenance and certificate extraction

The BANK invariant says that each deposit is justified, but Paper III needs more: if T appears at a finite stage, MAPQUEST must be able to exhibit the proof rather than merely remember that T was once accepted.

Accordingly, each noninitial deposit stores provenance

$$\text{prov}(C) = (r; A_1, \dots, A_k; \xi; \eta),$$

where r is the rule tag, $A_1, \dots, A_k$ are the exact premises, ξ is the finite instantiation/witness data, and η contains whatever certificate information the verifier requires.

Assume provenance is *backward-founded*: every premise pointer of a deposited conclusion refers either to an initial item or to an earlier deposit. In a finite run, following such pointers can therefore never produce an infinite descending chain.

Lemma 2 (Finite Target Provenance). *If $T \in B_m$ for some finite stage m and every noninitial deposit has finite backward-founded provenance, then the transitive provenance closure of T is a finite directed acyclic graph D_T whose leaves lie in B_0 .*

Proof. Only finitely many deposits occur before stage m . Every provenance edge points to an earlier stage or to an initial BANK entry. Therefore the subgraph reachable by tracing premise pointers backward from T is contained in a finite set and contains no directed cycle. Its leaves have no prior deposit provenance and hence lie in B_0 . $\square$

A topological ordering of D_T yields a finite ordered certificate

$$\Pi_T = (c_1, c_2, \dots, c_n)$$

whose final conclusion is exactly T .

8 The climax: MAPQUEST Target-Certification Theorem

MAPQUEST Target-Certification Theorem

Theorem 3 (MAPQUEST Target-Certification Theorem). *Let*

$$\mathfrak{F} = (\Sigma, \Gamma, \mathcal{R}, V)$$

be a frozen formal environment. Assume:

- (H1) $B_0 = \Gamma \cup \text{Ax}(\mathfrak{F})$;
- (H2) *every noninitial BANK deposit is the conclusion of a finite legal inference whose premises already lie in the BANK;*
- (H3) *every such deposited inference is accepted by a total sound verifier V ;*
- (H4) *every noninitial BANK entry retains finite backward-founded provenance sufficient to reconstruct the accepted inference;*
- (H5) *a finite MAPQUEST run reaches the fixed target: $T \in B_m$ for some $m < \infty$.*

Then MAPQUEST can extract a finite certificate Π_T such that

$$V(\Gamma, T, \Pi_T) = 1.$$

Consequently,

$$\Gamma \vdash_{\mathfrak{F}} T.$$

Proof. By the Finite Target Provenance Lemma, tracing the stored provenance of T backward yields a finite DAG D_T whose leaves lie in $B_0 = \Gamma \cup \text{Ax}(\mathfrak{F})$ and whose internal nodes are verifier-accepted conclusions of legal inferences from earlier nodes.

Choose a topological ordering of the internal nodes. Reading the associated stored rule tags, premises, witness data, and local certificate data in that order produces a finite certificate

$$\Pi_T = (c_1, \dots, c_n)$$

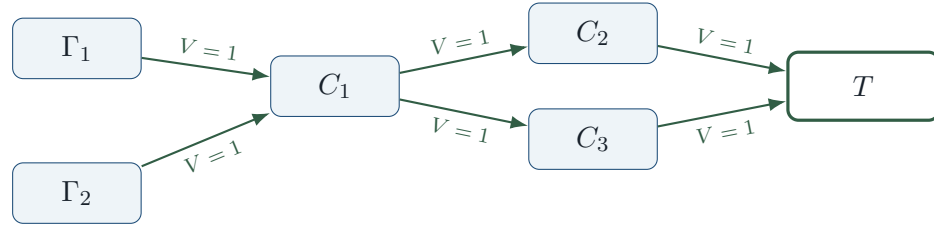
ending in T . By hypothesis, every constituent inference was accepted by the same declared verifier and its premises occur earlier in the ordered certificate or in B_0 . Therefore replaying the extracted certificate succeeds:

$$V(\Gamma, T, \Pi_T) = 1.$$

Soundness of V then implies

$$\Gamma \vdash_{\mathfrak{F}} T.$$

□



The search history may contain
millions of rejected candidates.
The extracted proof keeps only the finite
verifier-accepted dependency DAG ending at T .

Figure 2: Target certification extracts the accepted dependency DAG; rejected search events are not proof steps.

9 Why this is the theorem we wanted

The reminder at the beginning of the paper was

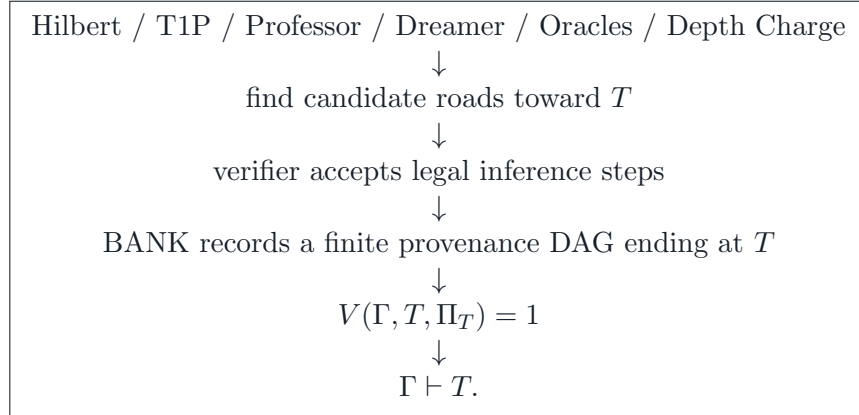
the ATP, or theorem-proving AMLD, is trying to prove T .

The Target-Certification Theorem does exactly that for a completed successful run. Its hypotheses do not assert in advance that $\Gamma \vdash T$. They assert operational facts about the environment, the verifier, BANK admission, provenance, and the concrete terminal event $T \in B_m$.

From those mild conditions the theorem derives the desired statement:

$$\Gamma \vdash_{\mathfrak{F}} T.$$

The navigation machinery is therefore placed in its proper logical role:



What this theorem does not say. It does not say that every arbitrary target T must be reached in finite time. Paper II gives positive completeness conditional on the existence of a finite proof and fair effective search. Paper III says something different: once a finite verifier-gated run really reaches T , the recorded route is enough to certify that T has been proved.

10 When MAPQUEST says it reached T , did it actually prove T ?

We can now ask the question in its final form.

When MAPQUEST says it reached T , did it actually prove T ?

If “reached” means merely that a heuristic score became small, that a Hilbert coordinate approached a target region, that the Professor assigned high partial credit, or that Dreamer proposed a plausible route, the answer is *no*. Those are search events.

If instead “reached” means that T entered the trusted BANK through a finite sequence of provenance-preserving, verifier-accepted deposits, then the Target-Certification Theorem applies. MAPQUEST extracts Π_T , replays it under the declared verifier, and obtains

$$V(\Gamma, T, \Pi_T) = 1.$$

Therefore

$$\Gamma \vdash_{\mathfrak{F}} T.$$

Yes.

MAPQUEST does not prove T because its probe reached a favorable Hilbert address.

It proves T because the navigation machinery constructed a finite chain of legal, verifier-accepted inferences whose terminal formula is exactly T .

11 Conclusion

Paper I built the MAPQUEST navigator. Paper II ensured that explicit addressability and Hilbert-fair recurrence can preserve access to every existing finite proof route under the declared effective assumptions. Paper III closes the logical loop.

The search can be elaborate. The proof criterion is simple:

$$\boxed{\Gamma \xRightarrow[V=1]{c_1} C_1 \xRightarrow[V=1]{c_2} \cdots \xRightarrow[V=1]{c_n} T.}$$

When the finite accepted route is extractable and the verifier accepts the assembled certificate,

$$\boxed{V(\Gamma, T, \Pi_T) = 1 \implies \Gamma \vdash T.}$$

That is the target-certification event toward which the preceding MAPQUEST machinery was built.